On Gated Permutations and Optimal Non-Collapsing Hash Functions

Joint work with Joseph Carolan and Vinod Vaikuntanathan

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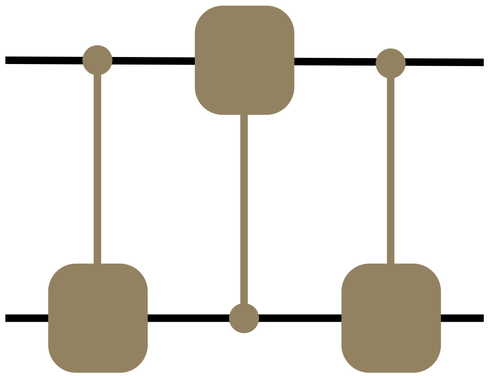

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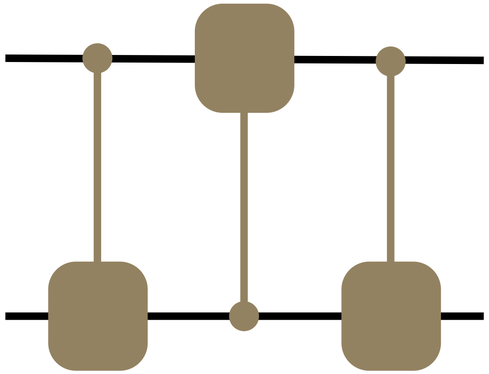

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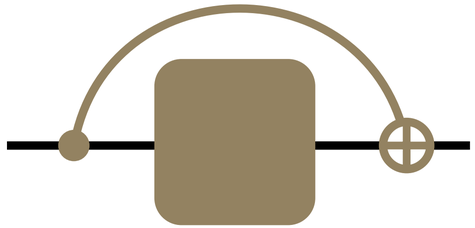

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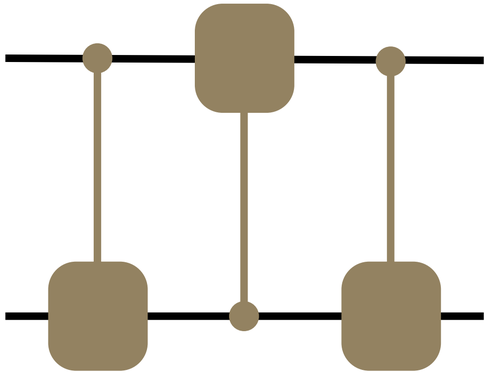

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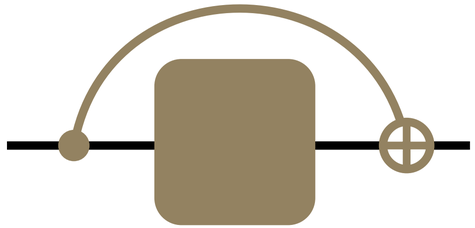

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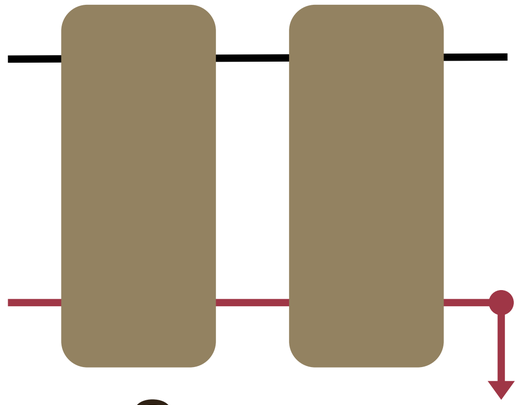

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- The work of [Car25], building on [ACMT25], shows how to analyze random permutations via the *compressed permutation oracle*

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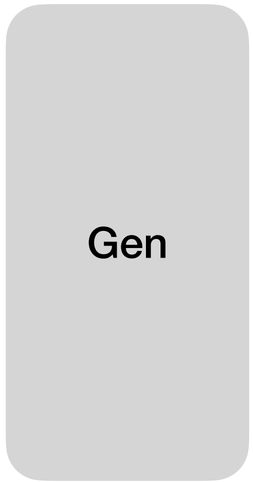

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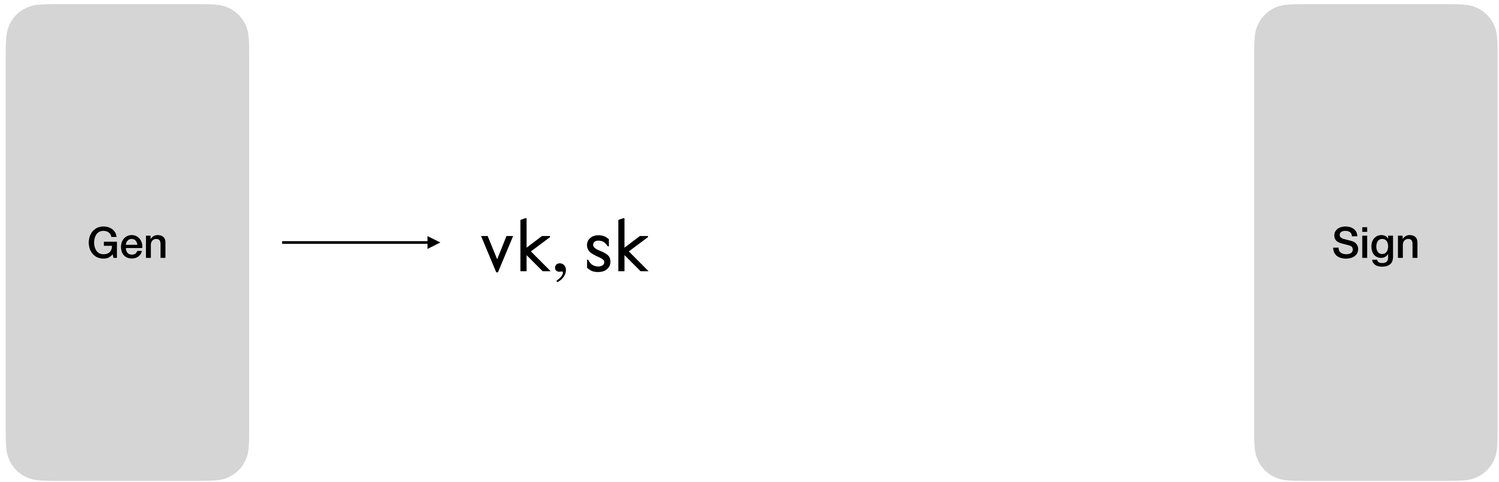

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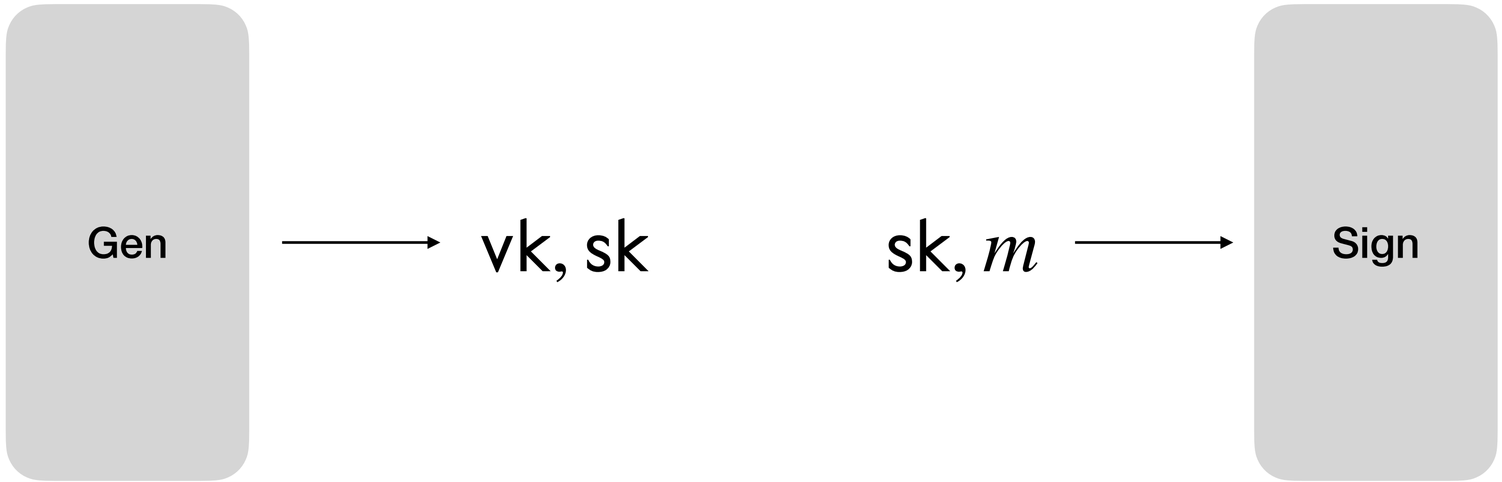

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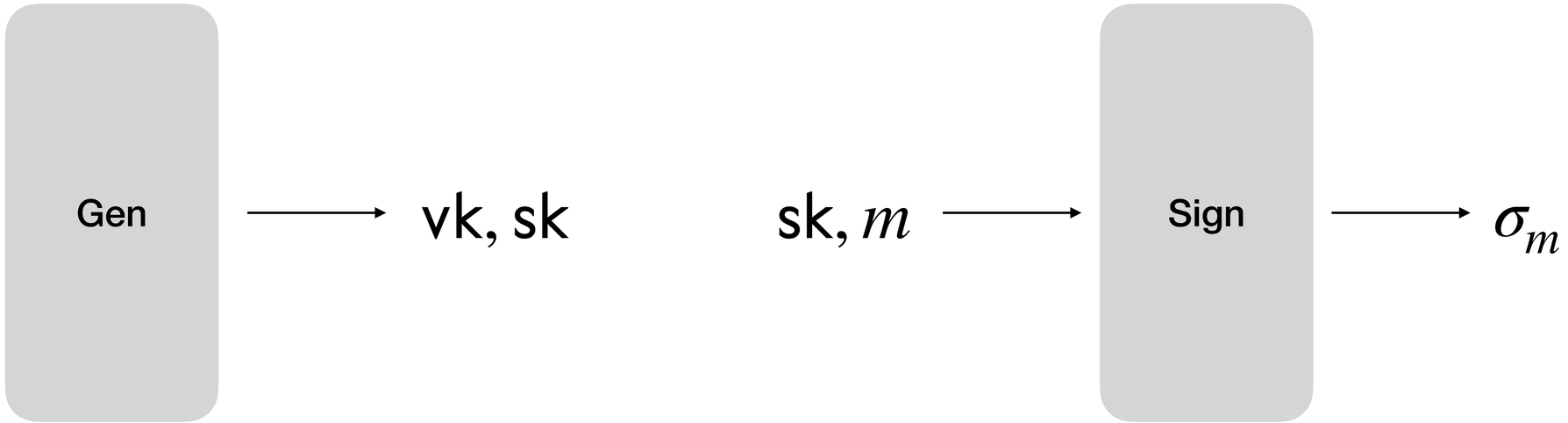

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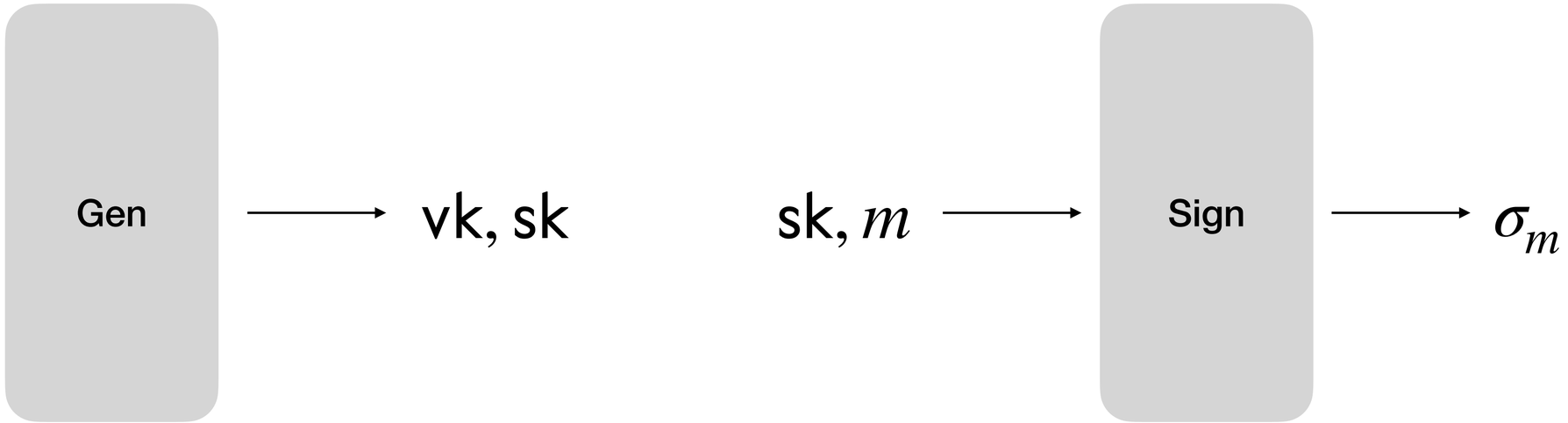

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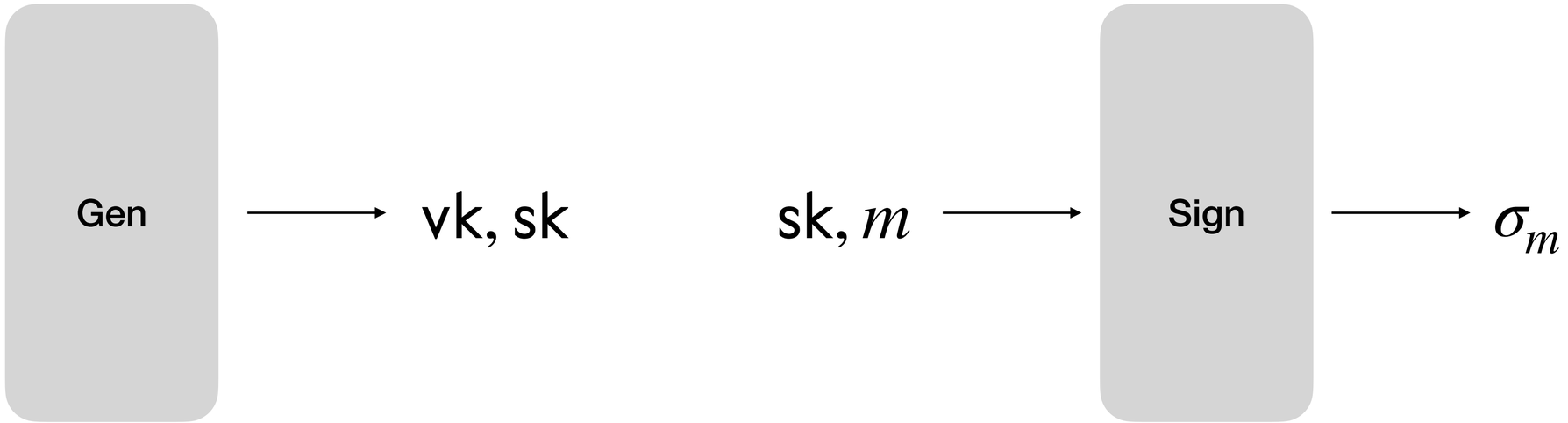

- Verification is **public**: **anyone** with the verification key can verify
- **Security**: *only the owner of the signing key* can produce valid signatures

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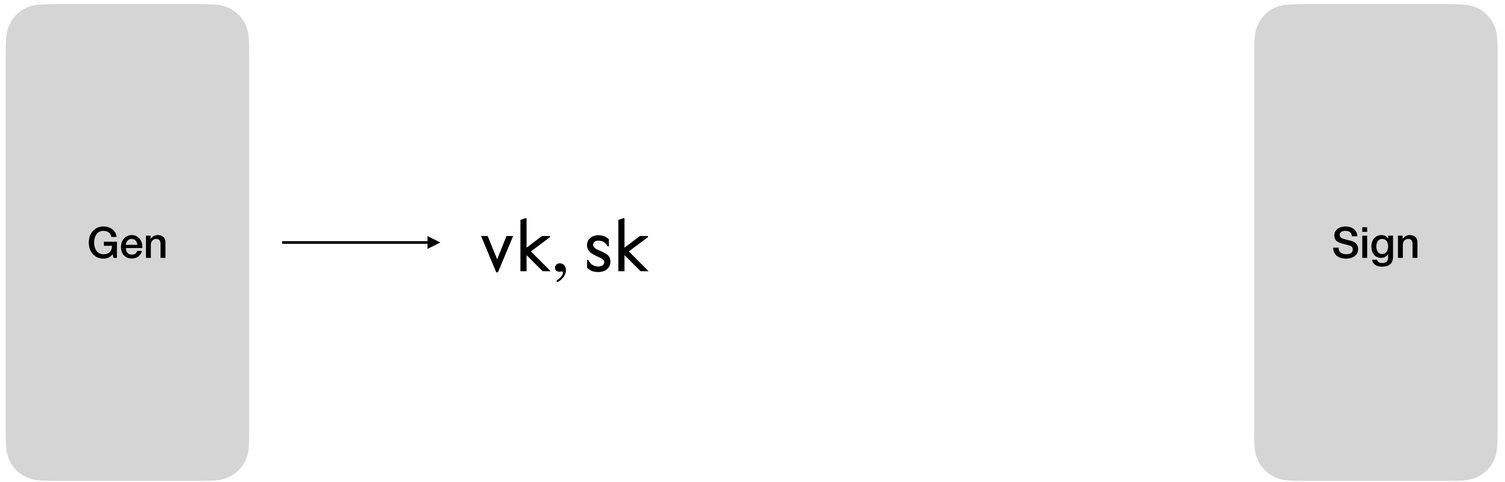

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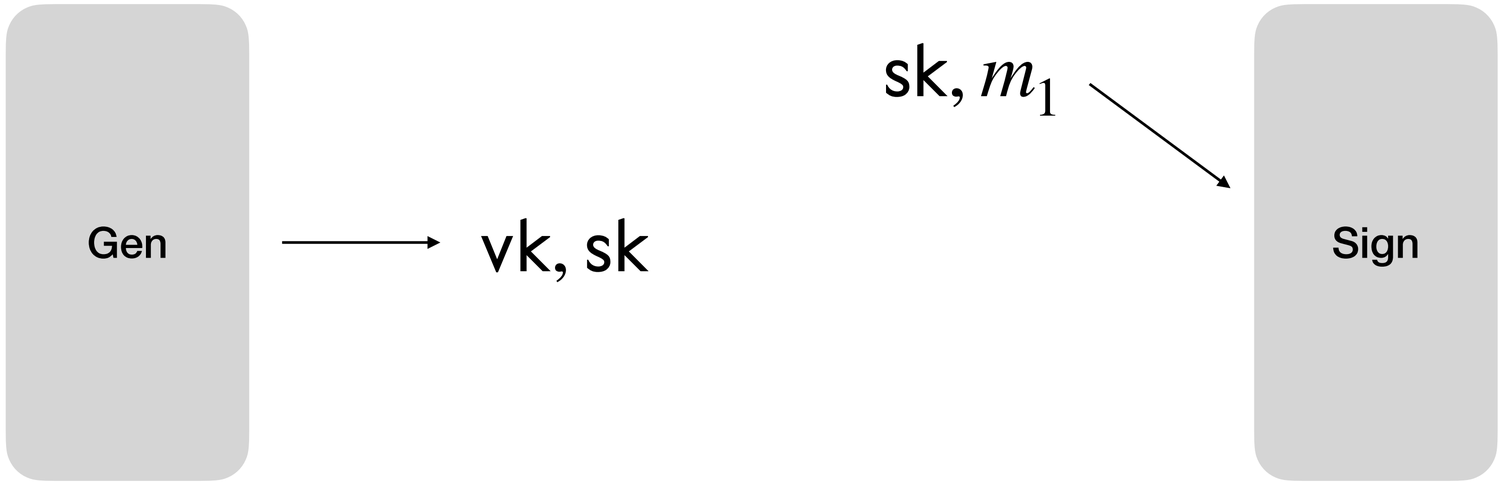

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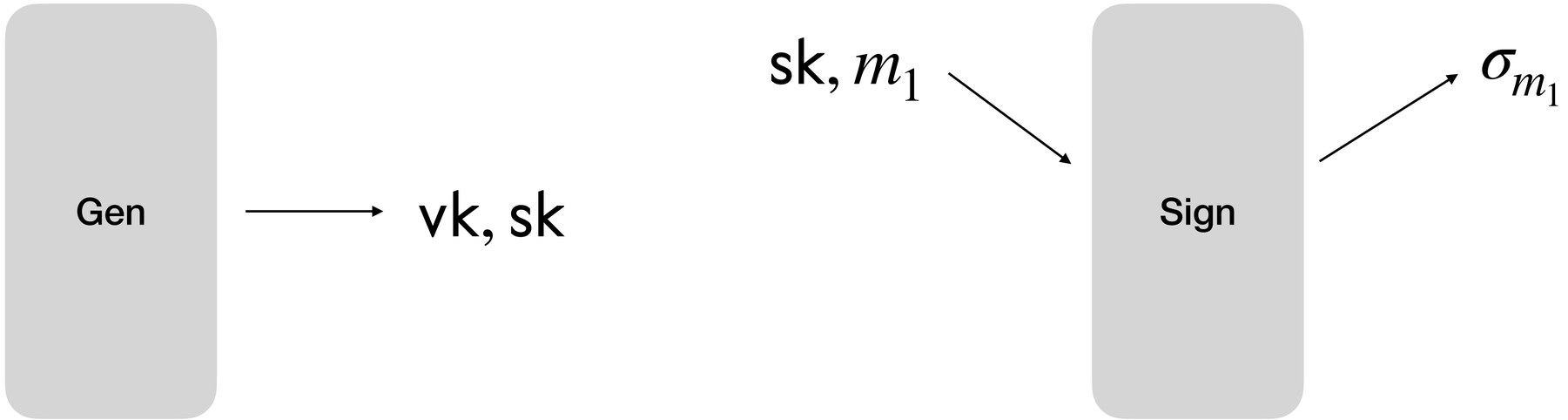

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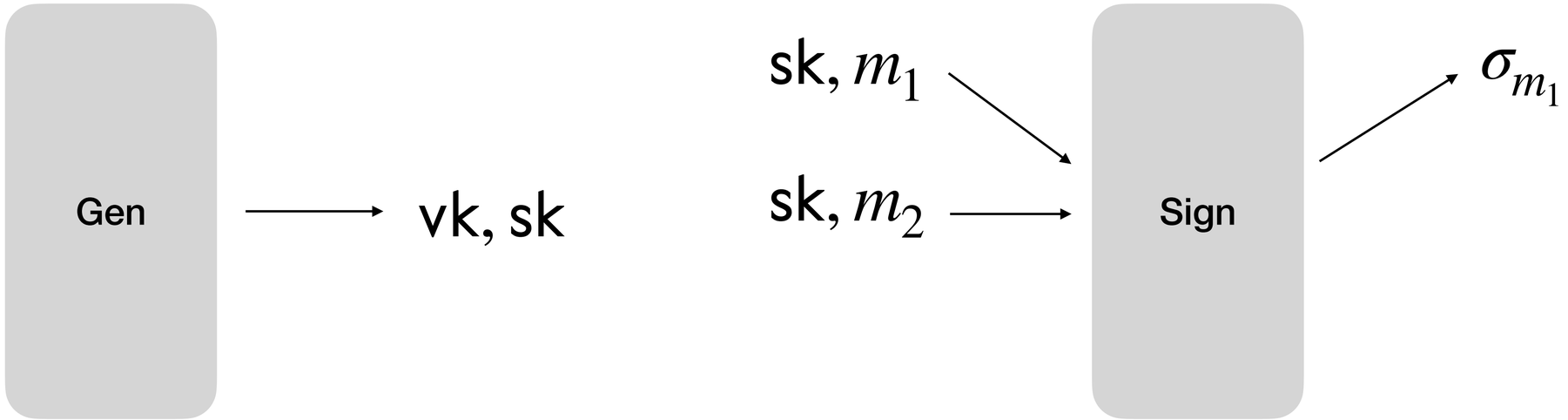

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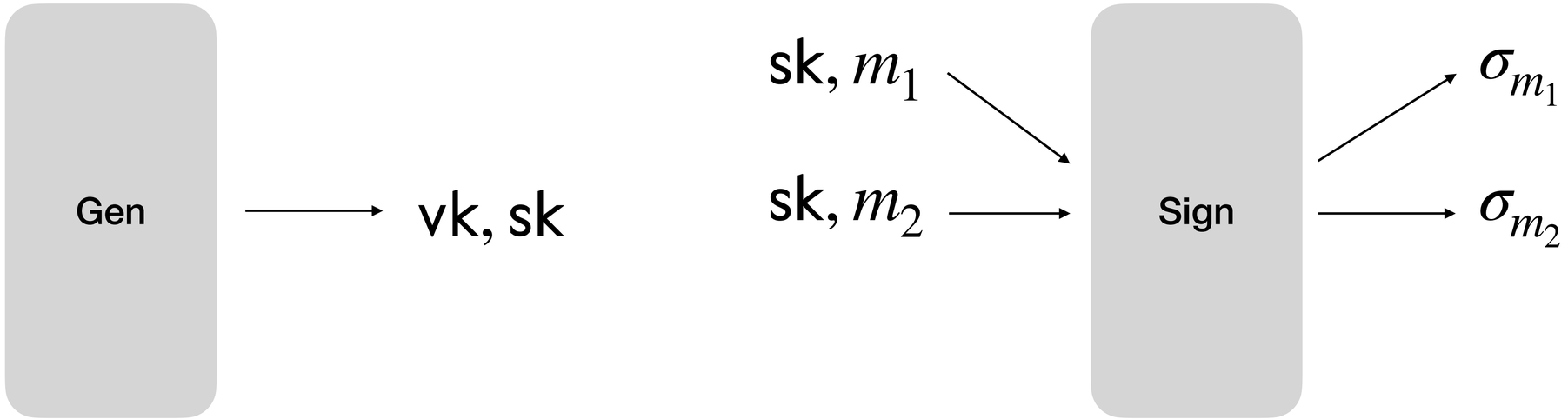

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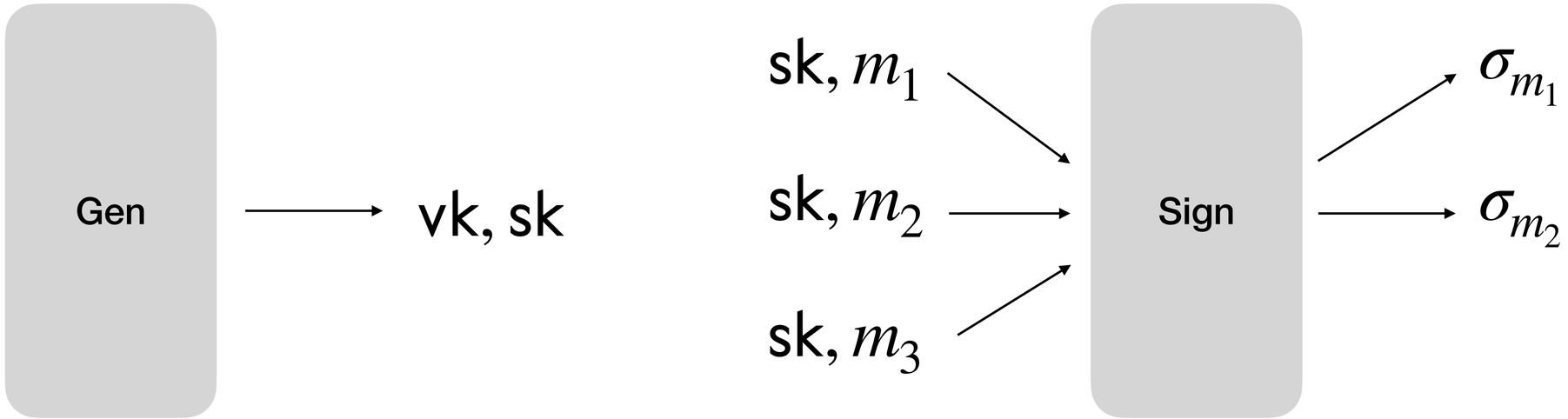

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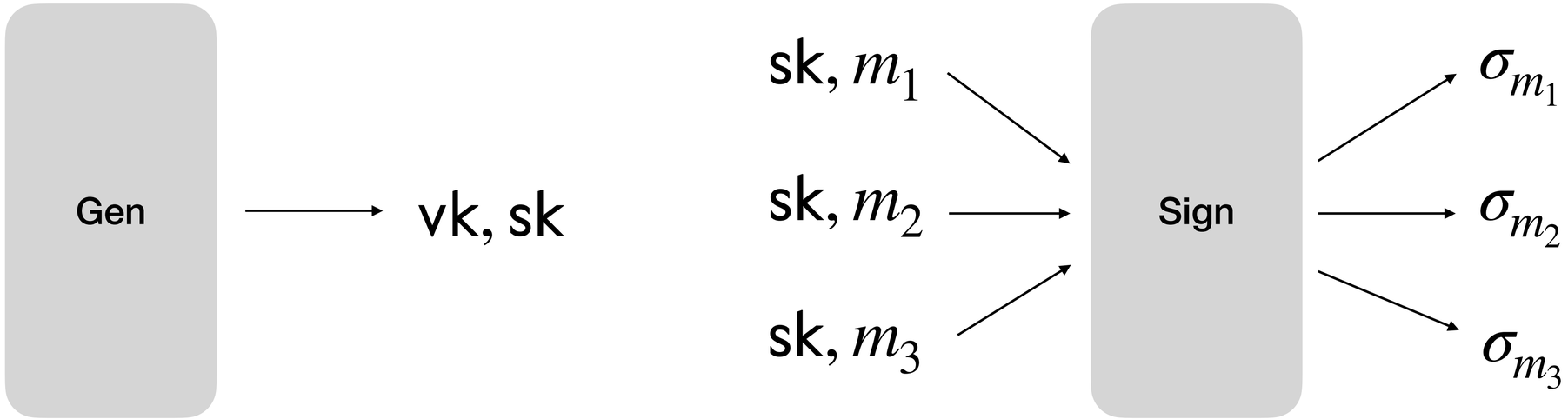

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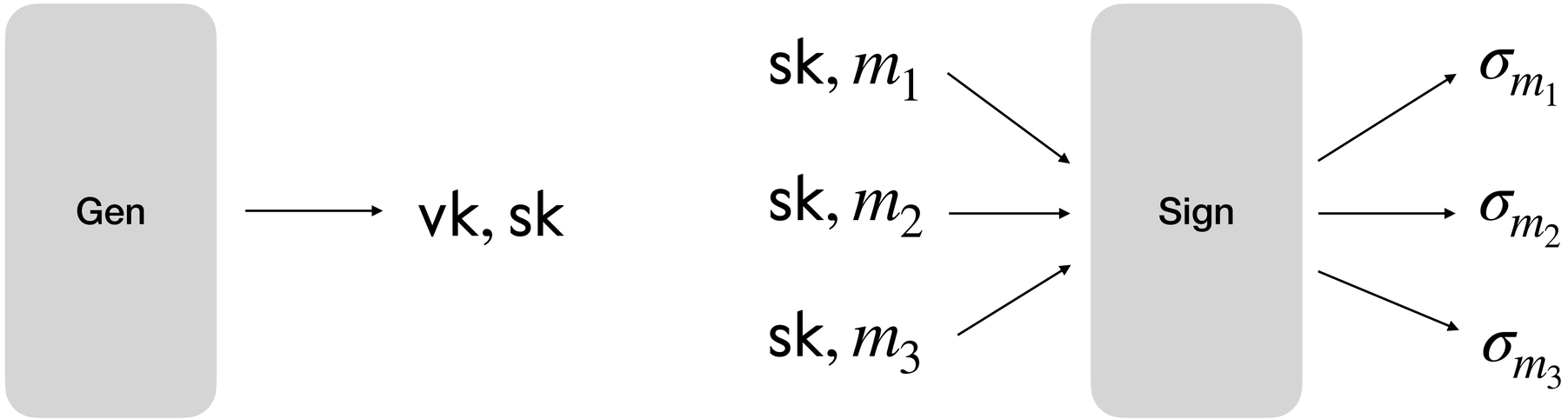

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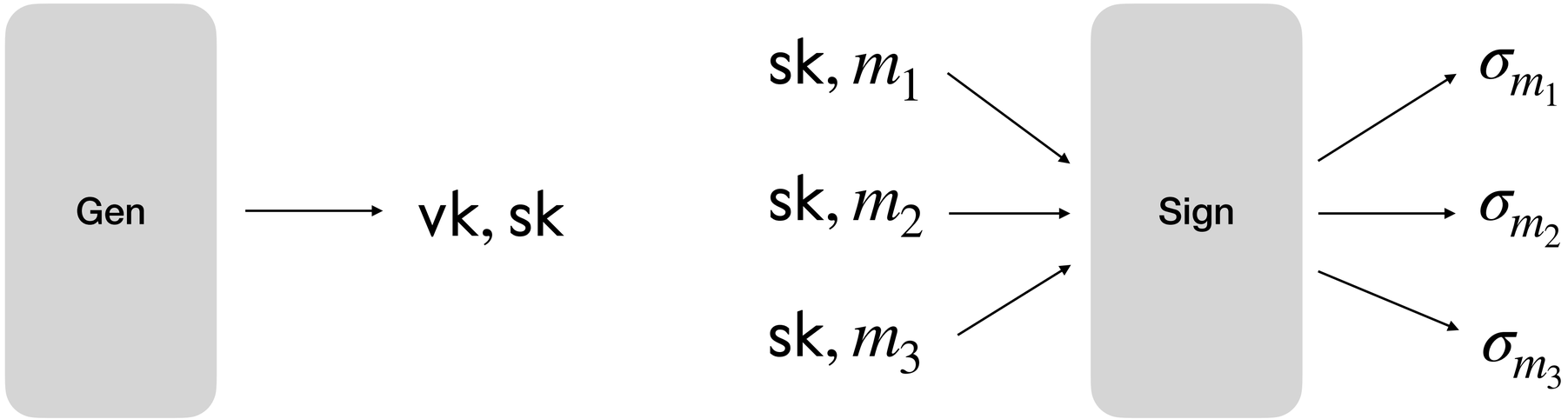

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- Clearly impossible in the classical model…

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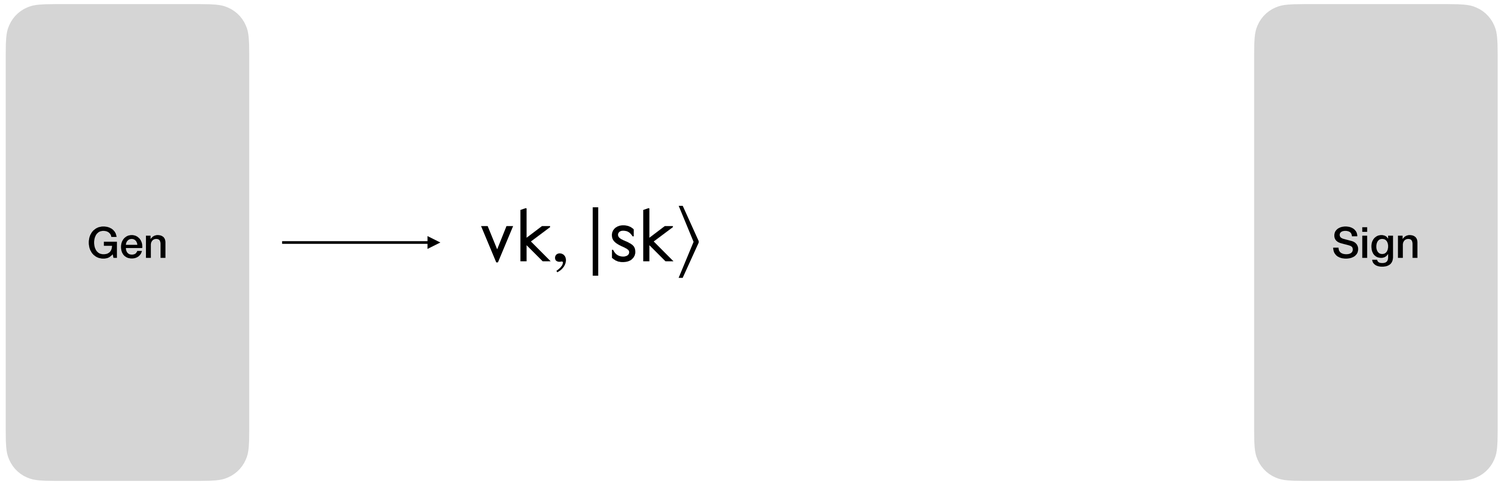

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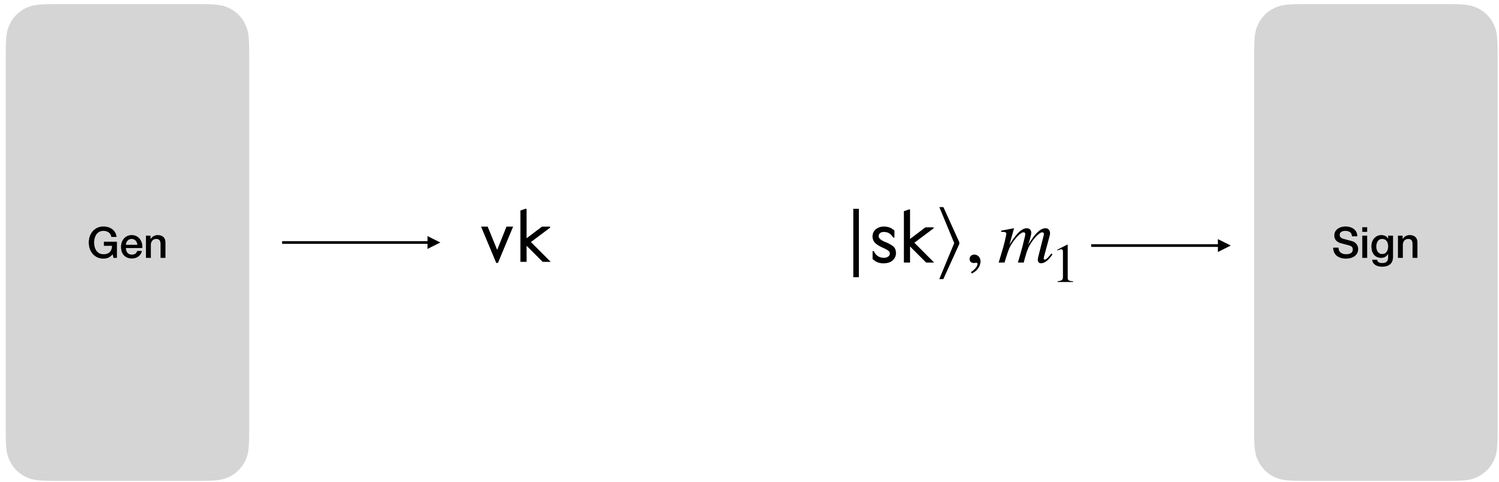

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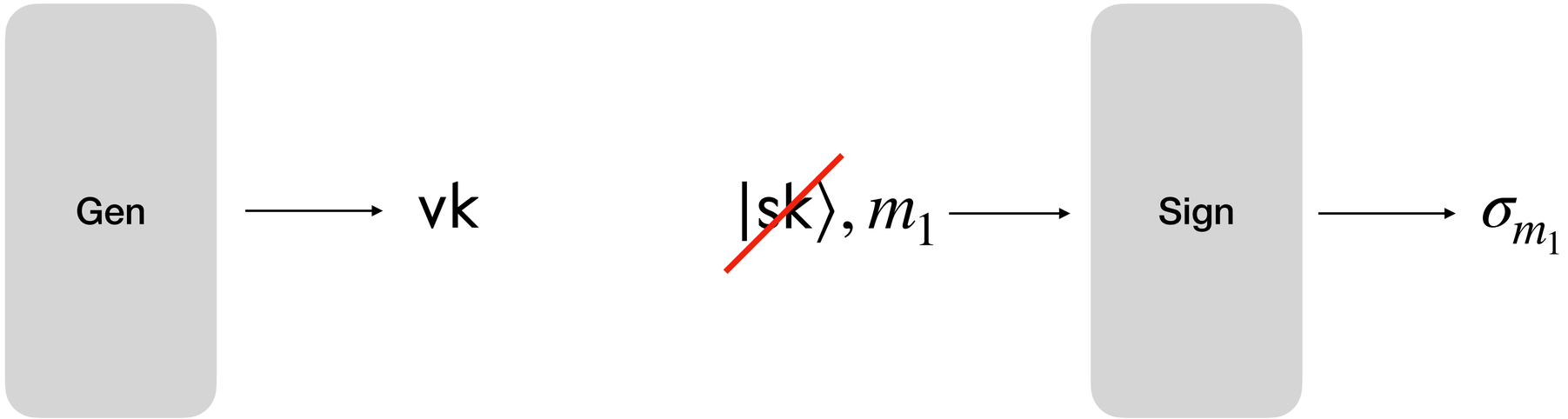

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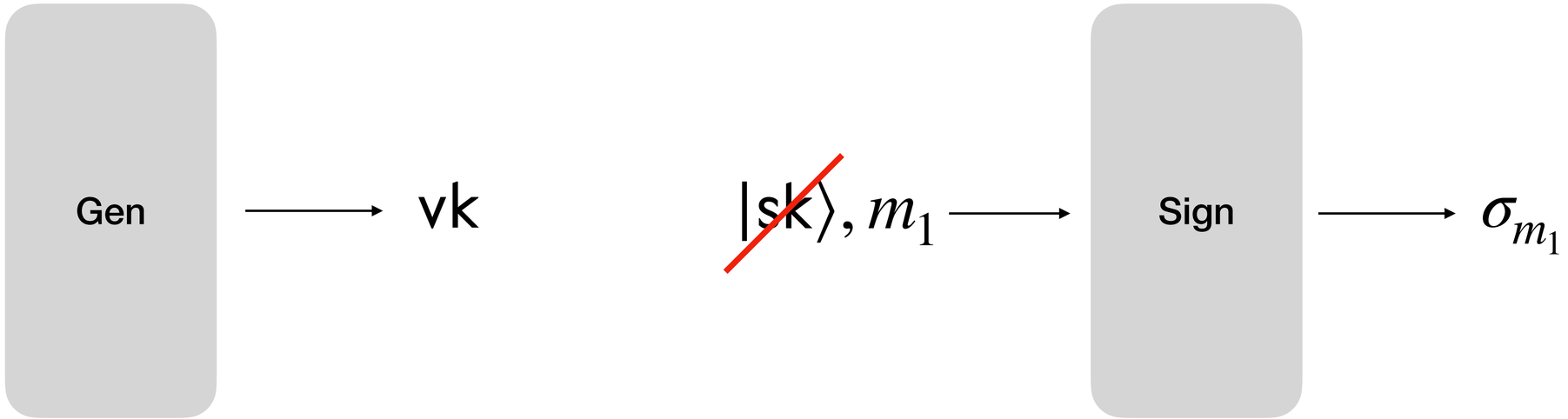

- Implies quantum lightning, quantum money with classical communication, proofs of quantumness, certified randomness, …

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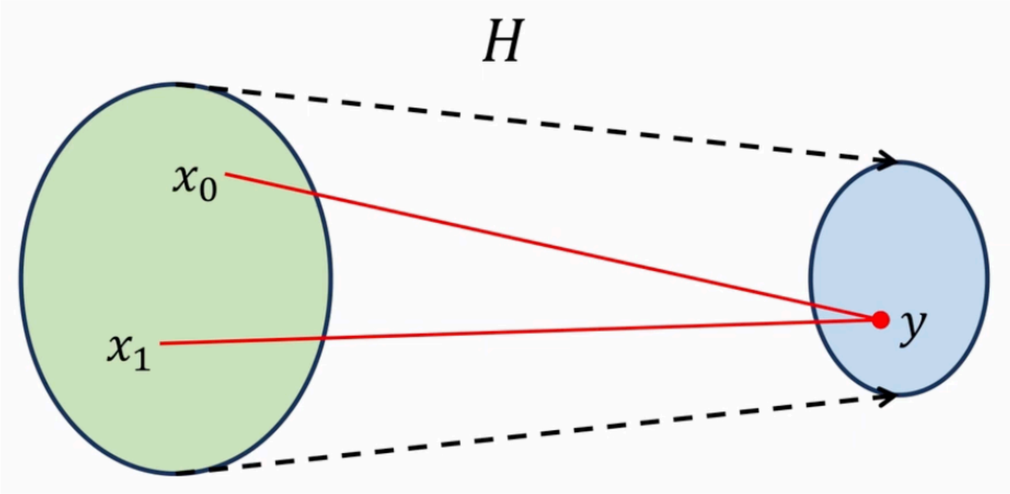

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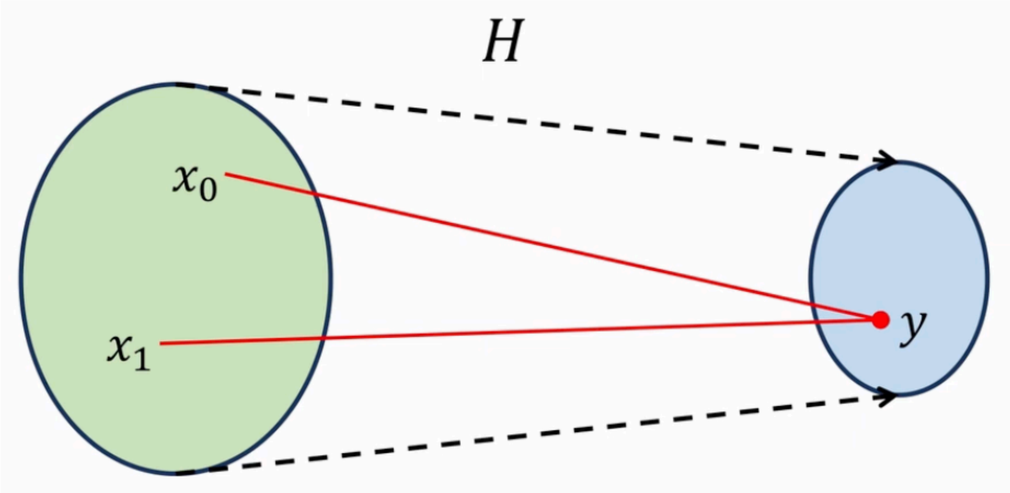

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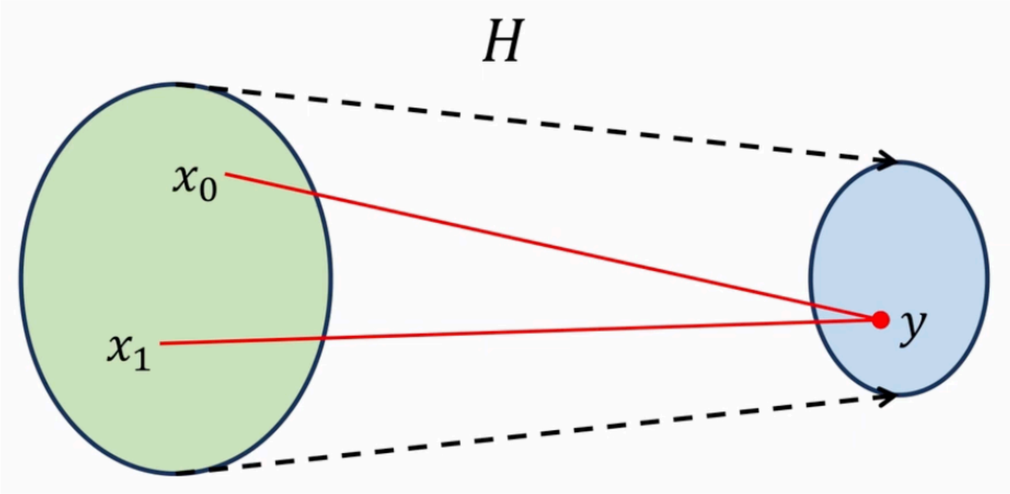

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- Quantumly, this is not known to be equivalent!

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- In theory, it might be able to transform $|H^{-1}(y)\rangle$ to get preimage of its choosing

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- [Zha17, AGKZ20, DS22]: non-collapsing CRHF $\Longrightarrow$ one-shot signature!

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- **Problem:** H_π is not even classically one-way when given access to π^{-1}!
- Given $y \in \{0,1\}^r$, an adversary can choose any $z \in \{0,1\}^{n-r}$ and compute a preimage $x = \pi^{-1}(y \| z)$.

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- **Bad news:** this construction is actually collapsing!

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- We show that when each $F(y, \cdot)$ is independently drawn from a family $\mathscr{F}$ with mild statistical properties (e.g. bounded preimages/pairwise independent), then H is one-way/collision-resistant when given access to $(G_{\pi,F}, G^{-1}_{\pi,F})$

Proof Sketch

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- Let's think about the proof in the *classical* setting, which begins by simulating $G_{\pi,F}$ on-the-fly via *lazy sampling*

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- Let's think about the proof in the *classical* setting, which begins by simulating $G_{\pi,F}$ on-the-fly via *lazy sampling*
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- Whenever a new query is made, we can sample a new input-output pair, provide these values, and update D

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- Combining our two bounds gives an upper bound of roughly $O(q \cdot (2^{-r} + p_1))$

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- Solution: Purify random coins r in F and coherently track consistent pairs (r, z)

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- If we instantiate $\mathscr{F}$ appropriately, it is simple to construct a collapsing distinguisher [SZ25, HSVZ26]
- Applications: optimal/tight constructions of non-collapsing CRHFs, one-shot signatures, and quantum fire schemes in the classical oracle model
- Also: more general version of the fundamental lemma which relates an algorithm's output to the values recorded in the compressed database